

process_options

```
log_action_begin_msg "Reconfiguring network interfaces"
ifdown -a --exclude=lo || true
ifup -a --exclude=lo; then
    log_action_end_msg $?
else
    log_action_end_msg $?
fi
::
<<<code truncated>>>
```

Now, let's focus on another important command that we've discussed previously—*ifconfig*. In the last issue we looked at how to obtain details like the MAC address and so on, about the network interfaces. In this edition, we will explore the ways to use this command.

- *up*: This can be used to activate a particular interface
- *down*: As the name suggests, you may use this to deactivate an interface (in order to obtain the list of the interface, you can issue the *ifconfig* command)
- *arp*: If you wish to enable (or disable) the address resolution protocol on a particular interface, you can try this
- *allmulti*: This is another option that is used in the intranet zones of many organisations. The main use of this command is to enable (or disable) the reception of any multicast packets. You may employ this if you wish to send multicast messages to a set of hosts with special network addresses (destination hosts). A typical example could be the case where a university is multicasting special lectures to students of a particular department who are connected to the university intranet.
- *netmask <address>*: This helps you to assign a network mask of the network to which you want to connect a system. I suggest you explore topics like DHCP and auto-configure mechanisms to learn more about typical implementations in modern operating systems.
- *mtu N*: If you wish to set a custom maximum transmission unit (MTU) value, you can try this. Different network types recommend different network values. Here are a few of them (format used – *Network: MTU in bytes*):
 - Ethernet: 1500
 - IEEE 802.3/802.2: 1492
 - 16 Mbit/Sec Token Ring: 17914
 - 4Mbits/Sec Token Ring: 4464
 - FDDI: 4352

In my Debian-based system, I can issue a command like the one below to change the MTU values:

```
sudo ifconfig eth0 mtu 1500
```

- *broadcast <address>*: You can enable (or disable) the interface to accept datagrams that are carrying the broadcast address.

Configuration*

As with the standard resolver library, the file */etc/resolv.conf* must be set up before the resolver can function. In addition, the file */etc/host.conf* contains configuration information specific to *resolv*.

The *host.conf* file should contain one configuration keyword per line, followed by appropriate configuration information. The keywords recognised are *order*, *trim*, *multi*, *nospoof*, and *reorder*. Each keyword is described below:

- *order*: This keyword specifies how host lookups are to be performed. It should be followed by one or more lookup methods, and separated by commas. Valid methods are *bind*, *hosts* and *nis*.
- *trim*: This keyword may be listed more than once. Each time it should be followed by a single domain name, with the leading dot. When set, the *resolv*+ library will automatically trim the given domain name from the end of any hostname resolved via DNS. This is intended for use with local hosts and domains. (Note: *trim* will not affect hostnames gathered via NIS or the *hosts* file. Care should be taken to insure that the first hostname for each entry in the *hosts* file is fully qualified or non-qualified, as appropriate for the local installation.)
- *multi*: Valid values are *On* and *Off*. If set to *On*, the *resolv*+ library will return all valid addresses for a host that appears in the */etc/hosts* file, instead of only the first. This is, by default, as it may cause a substantial performance loss at sites with large *hosts* files.
- *nospoof*: Valid values are *On* and *Off*. If set to *On*, the *resolv*+ library will attempt to prevent hostname spoofing to enhance the security of *rlogin* and *rsh*. It works as follows: after performing a host address lookup, *resolv*+ will perform a hostname lookup for that address. If the two hostnames do not match, the query will fail.
- *alert*: If this option is set to *On* and the *nospoof* option is also set, *resolv*+ will log a warning of the error via the *syslog* facility. The default value is *Off*.
- *reorder*: Valid values are *On* and *Off*. If set to *On*, *resolv*+ will attempt to reorder host addresses so that local addresses (that is, on the same subnet) are listed first when a *gethostbyname*() is performed. Reordering is done for all lookup methods. The default value is *Off*.

*Adapted from *resolv*+ man page.

- *irq <address>*: This command is not used frequently. But you can experiment by setting the IRQ of the hardware, using this command.
 - *pointtopoint <address>*: You can set the address of the machine at the remote end of a point-to-point link.
 - *hw <type> <address>*: This allows you to assign the hardware (hw) address of devices. You may Google for "Change MAC address in Linux" or "GNU MAC Changer" to get more information. (Please be very careful while using this command, especially when you are in a VPN or in a network that uses special authentication methods.)
- Last month we discussed the *ethool* and its uses